Name: ______________

Date: ______________

Polynomial Functions

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can predict the end behavior of a polynomial from its degree and leading coefficient.**

RELATED SKILLS

- I — Infer polynomial end behavior from degree and leading coefficient.
- R — Identify terms, factors, coefficients, variables, and degree in a polynomial expression.

☐ **2. I can determine polynomial zeros and their multiplicities.**

RELATED SKILLS

- I — Relate zero multiplicity to local graph behavior.
- D — Determine real and complex zeros using an appropriate method.
- A — Identify terms, factors, coefficients, variables, and degree in a polynomial expression.

☐ **3. I can construct a polynomial function from specified zeros or graphical features.**

RELATED SKILLS

- D — Build a polynomial from zeros, multiplicities, and a point.
- A — Determine real and complex zeros using an appropriate method.
- A — Relate zero multiplicity to local graph behavior.

☐ **4. I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.**

RELATED SKILLS

- D — Infer polynomial end behavior from degree and leading coefficient.
- D — Relate zero multiplicity to local graph behavior.
- A — Determine real and complex zeros using an appropriate method.

☐ **5. I can determine the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.**

RELATED SKILLS

- I — Use inflection points to establish a minimum possible polynomial degree.
- I — Use turning points to establish a minimum possible polynomial degree.
- I — Use zeros and their multiplicities to establish a minimum possible polynomial degree.
- A — Infer polynomial end behavior from degree and leading coefficient.

☐ **6. I can build or select a polynomial model for a context and interpret its features.**

RELATED SKILLS

- D — Explain a model parameter in the language of its context.
- D — Select a function family whose behavior fits a table, graph, or context.
- A — Build a polynomial from zeros, multiplicities, and a point.
- A — Restrict a model to meaningful contextual inputs.